No :

H/W

① $\dfrac{8.765 \times \sqrt[3]{27.03}}{24.51}$

$x = \dfrac{8.765 \times \sqrt[3]{27.03}}{24.51} \rightarrow lg\left(\dfrac{8.765 \times \sqrt[3]{27.03}}{24.51}\right)$

$lg\, x =$

$lg\, x = lg\, 8.765 + lg\, 27.03^{1/3} - lg\, 24.51$

$lg\, x = lg\, 8.765 + lg\, \frac{1}{3} \times 27.03 - lg\, 24.51$

~~$lg\, x = 0.9427 + 9.01 - 3894$~~

~~$lg\, x = \text{Anti } lg\ 9.9527 - 38.94$~~

$lg\, x = lg\, 8.765 + lg\, 9.01 - lg\, 24.51$

$lg\, x = 0.9427 + 0.9547 - 1.3894$

$lg\, x = 0.4080$

$x = \text{Anti } log\ 0.4080$

$= 10^{0} \times 2.559$

$= 1 \times 2.559$

$= 2.559$

(9) $\sqrt{\frac{9.18 \times 8.02^2}{9.83}}$

$$\cancel{lg}\, x = \sqrt{\frac{9.18 \times 8.02^2}{9.83}}$$

$$lg\, x = lg\left(\sqrt{\frac{9.18 \times 8.02^2}{9.83}}\right)$$

$$lg\, x = lg\, 9.18^{\frac{1}{2}} + lg\, 8.02^2 - lg\, 9.83$$

$$lg\, x = lg\, \tfrac{1}{2} \times 9.18 + lg\, 2 \times 8.02 - lg\, 9.83$$

$$lg\, x = lg\, 4.59 + lg\, 2 \times 8.02 - lg\, 9.83$$

$$lg\, x = 0.6618 + 2 \times 0.9042 - 0.9926$$

$$lg\, x = 0.6618 + 1.8084 - 0.9926$$

$$lg\, x = 2.5502 - 0.9926$$

$$lg\, x = 1.6676$$

$$x = \text{Anti } lg\ 1.6676$$

$$= 10^1 \times 4.651$$

$$= 46.51$$

③ $\dfrac{\sqrt{0.0945} \times 4.821^2}{48.15}$

$x = \sqrt{\dfrac{0.0945 \times 4.821^2}{48.15}}$

$\lg x = \lg\left(\dfrac{\sqrt{0.0945} \times 4.821^2}{48.15}\right)$

$\lg x = \lg 0.0945^{1/2} + \lg 4.821^2 - \lg 48.15$

$\lg x = \lg \frac{1}{2} \times 0.0945 + \lg 2 \times 4.821 - \lg 48.15$

$\lg x = \lg 0.04725 + \lg 9.642 - \lg 48.15$

$\lg x = \bar{2}.6744 + 0.9842 - 1.6825$

$\lg x = \bar{1}.6586 - 1.6825$

$\lg x = \bar{3}.9761$

$x = \text{Anti} \lg \bar{3}.9761$

$= 10^{-3} \times 9.464$

$= 0.009464$

(iv) $\therefore \dfrac{3 \times 0.752^2}{\sqrt{\cancel{19}\ 17.96}}$

$$x = \frac{3 \times 0.752^2}{\sqrt{17.96}}$$

$$\lg x = \lg\left(\frac{3 \times 0.752^2}{17.96}\right)$$

$$\lg x = \lg 3 + \lg 0.752^2 - \lg 17.96$$

$$\lg x = \lg 3 + \lg 2 \times 0.752 - \lg 17.96$$

$$\lg x = \lg 3 + \lg 1.504 - \lg 17.96$$

$$\lg x = \cancel{1}\ 0.4771 + 0.0772 - 1.2544$$

$$\lg x = 0.5543 - 1.2544$$

$$\lg x = \bar{1}.2999$$

$$x = \text{Anti}\lg \bar{1}.2999$$

$$= 10^{-1} \times 1.994$$

$$= 0.1994.$$

① $\dfrac{6.591 \times \sqrt[3]{0.0782}}{0.9821^2}$

$$x = \frac{6.591 \times \sqrt[3]{0.0782}}{0.9821^2}$$

$$\lg x = \lg\left(\frac{6.591 \times \sqrt[3]{0.0782}}{0.9821^2}\right)$$

$$\lg x = \lg 6.591 + \lg 0.0782^{1/3} - \lg \overset{\cancel{0.8}}{\cancel{0.8}}\, 0.9821^2$$

$$\lg x = \lg 6.591 + \tfrac{1}{3} \times 0.0782 - \lg 2 \times 0.9821$$

$$\lg x = \lg 6.591 + \lg 0.0260 - \lg 1.9642$$

$$\lg x = 0.8190 + \bar{2}.4150 - 0.2932$$

$$\lg x = \bar{1}.2340 - 0.2932$$

$$\lg x = \bar{2}.9408$$

$$x = \text{Anti}\lg \bar{2}.9408$$

$$= 10^{-2} \times 8.726$$

$$= 0.08726$$

⑤ $\dfrac{3.251 \times \sqrt[3]{0.0234}}{0.8915}$

$$x = \frac{3.251 \times \sqrt[3]{0.0234}}{0.8915}$$

$$\lg x = \lg\left(\frac{3.251 \times \sqrt[3]{0.0234}}{0.8915}\right)$$

$$\lg x = \lg 3.251 + \lg 0.0234^{1/3} - \lg 0.8915$$

$$\lg x = \lg 3.251 + \lg \tfrac{1}{3} \times 0.0234 - \lg 0.8915$$

$$\lg x = \lg 3.251 + \lg 0.0078 - \lg 0.8915$$

$$\lg x = 0.5120 + \bar{3}.8921 - \bar{1}.9501$$

$$\lg x = \bar{2}.4041 - \bar{1}.9501$$

$$\lg x = \cancel{\bar{2}.3}\ \bar{2}.4540$$

$$x = \text{Anti}\lg \bar{2}.4540$$

$$= 10^{-2} \times 2.844$$

$$= 0.02844$$